

Sihan (Eric) Yan

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EDUCATION

University of Pennsylvania M.S.E. in Data Science • <i>Relevant Coursework:</i> Advanced Machine Learning, Advanced Deep Learning, Natural Language Processing, Big Data Analytics, Time Series Forecasting, Database and Information Systems	GPA: 3.8/4.0 Philadelphia, PA Aug 2024 – May 2026
Arizona State University B.S. in Business Data Analytics B.S. in Supply Chain Management • <i>Relevant Coursework:</i> Business Data Analytics, Business Database Systems, Data Modeling, Enterprise Analytics	GPA: 4.0/4.0 Tempe, AZ Aug 2020 – May 2024 Moeur Award, Dean's List

SKILLS

Languages: Python, SQL, R, JavaScript, HTML/CSS
Analytical: Regression, Time Series, A/B Testing, Hypothesis Testing, Correlation Analysis, KPI Reporting
Machine Learning: Random Forests, Logistic Regression, SMOTE, ROC/AUC, BERT embeddings, XGBoost
Tools: Tableau Certified, Power BI, Databricks, Airtable, Google Analytics, Salesforce, HubSpot, Qualtrics, Git
Databases: PostgreSQL, MySQL, MongoDB, Neo4j, 3NF Modeling, ERD Documentation

PROFESSIONAL WORK EXPERIENCE

University of Pennsylvania <i>Data Analyst, Alumni Marketing</i> • Normalized Penn EENT alumni engagement data from Airtable exports into an 11-table 3NF schema, reconciling Penn SIS and HubSpot CRM records across 6,833 contacts and 15,971 enrollment rows. • Built database documentation and handoff materials including ERD, data dictionary, table relationships, reference values, data quality issues, and recurring query use cases. • Automated 5 ETL pipelines integrating Qualtrics and HubSpot APIs, streamlining survey data processing and reducing manual workload by 80%.	Philadelphia, PA Aug 2024 – May 2025
St. Vincent de Paul <i>Data Scientist, Marketing Department</i> • Integrated paid media, Google Analytics, and Salesforce donation datasets across Google Grants, StackAdapt, Google Ads, Facebook, and Bing for channel performance reporting. • Built Tableau dashboards and Databricks workflows for campaign spend, sessions, conversions, and donation KPIs, reducing manual reporting time by 40%. • Automated SQL ETL workflows for fundraising data, increasing pipeline speed by 30% and supporting weekly reallocation of a USD 18,800 cross-channel media budget. • Built donor churn and retention models in Databricks using Random Forest, Logistic Regression, SMOTE, ROC/AUC, and confusion-matrix diagnostics.	Phoenix, AZ Aug 2023 – Apr 2024

PROJECT

News Source Classification NLP, Python • Built a 300k news headline dataset by scraping Fox and NBC with Selenium and BeautifulSoup, applying deduplication and text normalization to enable robust binary classification. • Built an NLP pipeline using BERT embeddings for news source classification, achieving 87% accuracy and outperforming baseline models by 12 percentage points.	Aug 2024 – Dec 2024
NBA Performance Analytics Platform Python, PostgreSQL • Cleaned and integrated 1.6M records from player statistics, shot logs, and game outcomes into a normalized PostgreSQL database with structured analytical tables. • Optimized SQL pipelines using CTEs and materialized views, reducing query latency from 22 seconds to under 1 second for repeatable performance analysis.	Sep 2024 – Dec 2024
Marketing Channel Regression Analysis Python, SQL, Salesforce • Modeled relationships among impressions, sessions, conversions, media spend, and donations across Google Ads, Google Grants, Bing Ads, and StackAdapt datasets. • Applied outlier handling, correlation analysis, R-squared, and MSE diagnostics to separate directional campaign signals from weak predictive relationships.	2024